

Instructions:

- Attempt all questions.
- Section 1: 5 MCQs (1 mark each).
- Section 2: 5 MSQs (1 mark each).
- Section 3: 5 Short Answer questions (2 marks each).
- Section 4: 2 Long Answer questions (5 marks each).

Section-1: Single Correct Option (MCQs)

1.1. If a language model $\mathcal{M}_1$ achieves a lower cross-entropy value than model $\mathcal{M}_2$ on a specific validation dataset, which of the following statements must be true?

- A. $\mathcal{M}_1$ will yield a lower perplexity score than $\mathcal{M}_2$ on that same validation dataset.
- B. $\mathcal{M}_1$ structurally utilizes a higher-order N-gram configuration than $\mathcal{M}_2$.
- C. $\mathcal{M}_1$ contains fewer non-zero parameters than $\mathcal{M}_2$.
- D. $\mathcal{M}_1$ is mathematically guaranteed to perform better on downstream sequence generation tasks.

1.1	/1
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1.2. In a Seq2Seq model trained for machine translation, teacher forcing refers to:

- A. Using a pre-trained encoder and freezing its weights
- B. Feeding the ground-truth target token at step t as input to the decoder at step $t + 1$ during training
- C. Feeding the model's own prediction at step t as input at step $t + 1$ during training
- D. Penalising the model when it predicts a token that is too long

1.2	/1
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1.3. Which model is trained by corrupting input text and reconstructing the original text using an encoder-decoder architecture?

- A. BERT
- B. GPT
- C. BART
- D. Word2Vec

1.3	/1
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1.4. In the translation prompt `Translate English to French: sea otter => loutre de mer, cheese =>`, the example `sea otter => loutre de mer` is used as:

- A. A reward signal
- B. A task demonstration
- C. A learned soft prompt
- D. A validation example

1.4	/1	
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1.5. Why is the dot product in self-attention scaled by $1/\sqrt{d_k}$ before applying the softmax?

- A. To ensure that the attention weights are always between -1 and $+1$.
- B. To speed up inference by reducing the number of multiply-accumulate operations.
- C. Because for large d_k , the dot products can grow in magnitude ($\propto \sqrt{d_k}$), pushing the softmax into saturation regions with near-zero gradients; scaling stabilizes training.
- D. To implement dropout on the attention scores.

1.5	/1	
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Section-2: Multiple Select Questions (MSQs)

2.1. Which of the following optimization objectives target or reflect the exact same mathematical criteria when training an N-gram language model?

- A. Maximizing the log-likelihood of the training text corpus.
- B. Minimizing the empirical cross-entropy on the training corpus.
- C. Maximizing the perplexity on the training corpus.
- D. Minimizing the Kullback-Leibler (KL) divergence between the empirical distribution of the training text and the model's distribution.

2.1	/1	
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2.2. Which of the following statements about the **vanilla Seq2Seq model (no attention)** are correct?

- A. The encoder and decoder can be different RNN architectures (e.g. LSTM encoder, GRU decoder).
- B. Performance typically degrades significantly for very long input sequences.
- C. The decoder can directly access any encoder hidden state at each time-step.
- D. Both encoder and decoder parameters are jointly trained end-to-end.
- E. The context vector dimension must equal the target vocabulary size.

2.2	/1	
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2.3. Which tasks are naturally suitable for BERT-style encoder models?

- A. Named Entity Recognition
- B. Sentiment classification
- C. Extractive question answering
- D. Unconditional story generation

E. Sentence-pair classification

2.3	/1	
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2.4. Which of the following describes Orca-style data generation?

- A. It improves the information content of responses.
- B. It uses diverse system instructions such as reasoning steps, CoT, ELI5, and helpful/informative responses.
- C. It is unrelated to instruction tuning.
- D. It can use teacher-model completions such as GPT-4/GPT-3.5 style outputs.

2.4	/1	
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2.5. Which of the following statements about **Reinforcement Learning for LLMs** are **correct**? (Select all that apply.)

- A. In the LLM-RL framework, the policy $\pi_\theta(a \mid s_t)$ is the LLM's softmax distribution over the vocabulary given the current context.
- B. The reward signal in standard RLHF is dense: the model receives a separate reward for every token it generates.
- C. The value function $V(s_t)$ is trained to predict the expected future cumulative reward from state s_t , and it is typically implemented as an additional linear head on the LLM's hidden states.
- D. PPO-Clip is an off-policy algorithm that allows multiple gradient updates on the same batch of samples by correcting for distribution shift with importance weights.

2.5	/1	
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Section-3: Short Answer Questions

3.1. In Katz Backoff, the formula for a bigram is:

$$P_{\text{Katz}}(w_i \mid w_{i-1}) = \begin{cases} P^*(w_i \mid w_{i-1}) & \text{if } c(w_{i-1}, w_i) > 0 \\ \alpha(w_{i-1})P_{\text{Katz}}(w_i) & \text{if } c(w_{i-1}, w_i) = 0 \end{cases}$$

Provide the exact algebraic expression for the backoff weight $\alpha(w_{i-1})$ that ensures the conditional distribution sums to 1.

3.1	/2	
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3.2. Explain the information bottleneck problem in vanilla Seq2Seq models. How does the attention mechanism address it? Describe the flow of information for a 4-token source sentence under both architectures.

3.2	/2	
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3.3. Define Masked Language Modeling and explain why it is suitable for bidirectional encoders.

3.3	/2	
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3.4. Given a soft prompt length m and embedding dimension d , write the formula for the number of trainable prompt-tuning parameters. Compute it for $m = 50$ and $d = 768$.

3.4	/2	
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3.5. The role of residual connections and layer normalization in Transformers.

- (a) State the mathematical form of the “Add & Norm” operation applied around the self-attention sub-layer. Why is this operation critical for training deep Transformer stacks?
- (b) Compare “pre-norm” and “post-norm” Transformer variants. In pre-norm, layer normalization is applied *before* the sub-layer: $\text{output} = x + f(\text{LayerNorm}(x))$. In post-norm, it is applied *after*: $\text{output} = \text{LayerNorm}(x + f(x))$. State one advantage of each variant for training.

3.5	/2	
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Section-4: Long Answer Questions

- 4.1. Explain the difference between Masked Language Modeling and autoregressive language modeling. Your answer should include context used, loss computation, one example, and suitable downstream tasks.

4.1	/5	
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4.2. Reward model training and regularized objective.

- (a) A reward model outputs the following scores for three preference pairs:

Pair	$r(x, y^+)$	$r(x, y^-)$
1	2.0	1.0
2	0.5	1.5
3	3.0	0.0

Compute the Bradley-Terry loss for each pair: $\ell_i = \log \sigma(r(x, y_i^+) - r(x, y_i^-))$, where $\sigma(z) = 1/(1 + e^{-z})$. Then compute the average loss $\bar{\ell} = -\frac{1}{3} \sum_i \ell_i$.

- (b) Write down the regularized RLHF reward maximization objective $\mathcal{J}(\theta)$ for a policy π_θ . For a single generated response y with reward $r(x, y) = 1.5$ and $\log(\pi_\theta(y | x)/\pi_{\text{ref}}(y | x)) = 0.3$, compute the regularized reward $r_s(x, y)$ with $\beta = 0.1$. Interpret whether this response is beneficial to keep or suppress.

4.2	/5	
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